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EMPIRICAL STUDY

To What Extent Are Multiword Sequences Associated With Oral Fluency?

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This study examined the relationship between oral fluency and use of multiword sequences (MWSs) across four proficiency levels (Low B1 to C1 of the Common European Framework of Reference). Data came from 56 learners taking the speaking test of the Test of English for Educational Purposes, and our analysis obtained different measures of fluency (speed, breakdown, repair) and MWSs (frequency, proportion, association). Results showed that (a) high-frequency n-grams correlated positively with articulation rate; (b) n-gram proportion correlated negatively with frequency of mid-clause pauses; and (c) n-gram association strength correlated positively with frequency of end-clause pauses and negatively with repair frequency. Qualitative analysis suggested that the test-takers borrowed some task-specific n-grams from the task instructions and used them frequently in their performance. Whereas lower proficiency speakers used these n-grams verbatim, C1 level speakers used them competently in a variety of forms. We discuss significant implications of the findings for phraseology and language testing research.

Keywords fluency; multiword sequences; proficiency level; n-grams

Introduction

Oral fluency, the "flow, continuity, automaticity, or smoothness of speech" (Koponen & Riggenbach, 2000, p. 6), is a prime characteristic of successful

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language communication that has recently become central to many studies of second language acquisition (SLA). Investigating fluency helps researchers to understand how a second language (L2) is processed, produced, and acquired. In addition, exploring fluency enables them to examine some abstract dimensions of SLA, including implicit knowledge, proceduralization of newly learned structures, and automatization of the processes of speech production (Loewen & Sato, 2018; Paradis, 2009; Suzuki & DeKeyser, 2017). For this reason, fluency is often examined in relation to other aspects of language processing and performance. The current study belongs to this body of research in aiming to investigate the relationship between fluency and use of formulaic language. The interest in this relationship is built on psycholinguistic research evidence suggesting that use of formulaic language reduces the amount of language planning and processing, and therefore facilitates oral production (see Siyanova-Chanturia & Van Lancker Sidtis, 2018, for a review).

One line of research in this area has focused on the use of *multiword sequences* (MWSs)—combinations of words that appear together highly frequently in a target language (Garner & Crossley, 2018)—and on its relationship with proficiency (e.g., Nattinger & DeCarrico, 1992; Pawley & Syder, 1983; Wood, 2015; Wray, 2002; see Granger, 2018, for a review). Findings suggest that more proficient L2 users have a better command of MWSs in terms of range, frequency, and sophistication (e.g., Durrant & Schmitt, 2009; Garner & Crossley, 2018; Kyle & Crossley, 2015; Paquot, 2019). The findings also indicate that use of MWSs is related to oral fluency (Boers, Eyckmans, Kappel, Stengers, & Demecheleer, 2006; Stengers, Boers, Housen, & Eyckmans, 2011; Tavakoli, 2011; Wood, 2009, 2010).

Although the existing research evidence highlights the relationship between fluency and MWSs, no study thus far has examined this relationship systematically across a range of assessed levels of proficiency to see if there is a linear relationship between the two. Understanding the relationship between MWSs and fluency at different levels of development is important for the study of SLA in shedding light on the processes of speech production. In addition, such knowledge can help the language testing discipline to develop a valid assessment of learner phraseological knowledge and its relationship to fluency at different levels of proficiency. The current study aimed to contribute to these areas by investigating, both quantitatively and qualitatively, the relationship between MWSs and fluency in 56 participants at four levels of proficiency assessed via the speaking paper of the Test of English for Educational Purposes (TEEP).

Background Literature

Multiword Sequences

Use of MWSs has recently attracted attention from SLA researchers because of its important role in language processing and production. "MWS" is a catch-all term referring to combinations of words that appear together highly frequently in a target language (Garner & Crossley, 2018; Jeong & Jiang, 2019). MWSs include various forms of lexical strings, such as idioms (e.g., *kick the bucket*, *spill the beans*), restricted collocations (e.g., *heavy traffic*, *blow a fuse*), phrasal verbs (e.g., *hung up*, *call off*), binomials (e.g., *bride and groom*, *ladies and gentlemen*), proverbs (e.g., *birds of a feather flock together*), and lexical bundles or *n-grams* as used in this study (e.g., *and so on*, *one of the*, *I don't know*). Since the advancement of corpus-based techniques, n-grams have received increased attention in SLA research. Unlike previous research relying on native-speaker intuition in describing language units, n-gram studies use an objective frequency-based approach to identify recurrent sequences of *n* (e.g., two, three, four, or more) consecutive words in learner corpora (Paquot & Granger, 2012).

This learner corpora research generally takes one of two approaches to analyzing MWS production. In the first approach, n-grams are measured using only text-internal data (i.e., learner corpora), by counting the number of contiguous strings of words of a given length. For example, Huang (2015) extracted recurrent strings of three to five words from L2 English argumentative essays written by two groups of first language (L1) Chinese learners, and compared them in terms of size, range, and accuracy (see Paquot & Granger, 2012, pp. 138–140, for other examples). In the second approach, researchers draw on text-external data—that is, reference corpora, such as the British National Corpus and the Corpus of Contemporary American English (COCA)—to extract lexical bundles (Garner & Crossley, 2018; Kyle & Crossley, 2015). For word combinations to qualify as lexical bundles, the combinations found in learner corpora (i.e., corpora of language produced by L2 learners) must appear in language used by L1 speakers. In this regard, the second approach is concerned with the extent to which L2 learners employ target like MWSs. In the current study, we adopted this second approach to defining and measuring MWSs, with a specific focus on bigrams and trigrams, based on earlier studies using COCA as a reference corpus (Garner & Crossley, 2018; Kyle & Crossley, 2015).

Oral Fluency

Fluency is often considered a complex and multifaceted construct that is difficult to define and measure (Lennon, 1990; Segalowitz, 2010). For the purpose of this study, we consider fluency to be the general ease, flow, and continuity of speech, characterized by temporal and acoustic features and by disfluency markers. Segalowitz (2010) defined fluency in terms of the three dimensions of cognitive, utterance, and perceived fluency. In this model, cognitive fluency refers to "the efficiency of operation of the underlying processes responsible for the production of utterances" and is distinguished from utterance fluency, "the features of utterances that reflect the speakers' cognitive fluency," and perceived fluency, "the inferences listeners make about speakers' cognitive fluency based on their perceptions" (Segalowitz, 2010, p. 165). As can be seen, utterance fluency is the only aspect of fluency in this model that can be measured objectively through the analysis of the acoustic characteristics of speech. Although the three dimensions of fluency are internally dependent and highly interrelated, in the current study we were interested in utterance fluency because it allows for an objective measurement of fluency.

Measuring fluency is complex for a range of theoretical and empirical reasons, including the difficulty of deciding which measures best characterize oral proficiency. Skehan (2003) and Tavakoli and Skehan (2005) proposed a triadic framework in which three distinct aspects of fluency are measured: (a) speed, that is, the flow and continuity of speech; (b) breakdown, that is, pauses and silences that break the flow of speech; and (c) repair, that is, monitoring and repair processes such as false starts and reformulations. Researchers have found the triadic framework useful in explaining different aspects of the speech production process. For example, speed fluency is hypothesized to reflect the degree of automaticity (Kahng, 2014; Skehan, 2014); breakdown fluency indicates difficulties at the conceptualization and formulation stages of Levelt's model (explained below; Kormos, 2006); and repair fluency reflects the monitoring processes of the L2 production process (Ahmadian, 2011; Kormos, 1999).

Multiword Sequences and Speech Production

Theoretical Accounts Explaining the Relationship

The important relationship between phraseological knowledge and oral proficiency has been explained in terms of psycholinguistic research evidence. Though it remains debatable whether MWSs are stored and retrieved as holistic units to the same extent as single-word items (Siyanova-Chanturia & Martinez, 2015), psycholinguistic research has suggested that MWSs (e.g., *in the middle of the*) are processed differently from infrequent strings of language (e.g., *in the front of the*), with processing advantages for the former over the latter in both receptive and productive language tasks (see Siyanova-Chanturia & Van Lancker Sidtis, 2018, for a review). The increase in speed of language processing, an advantage that enables speakers to produce utterances more fluently, is linked to freeing up the attentional resources that speakers need in order to attend to other aspects of language production such as articulation and monitoring (Kormos, 2006; Skehan, 2009). Earlier research has suggested that for L2 learners MWSs provide quick and cost-effective access to lexically and morphosyntactically correct language, allowing them to produce language that is beyond the capacity of the learners to generate or manipulate creatively (Myles, Hooper, & Mitchell, 1998).

Levelt's (1989) speech production model, initially proposed for L1 speakers and later refined to cover L2 speakers (Kormos, 2006), is helpful in explaining the relationship between use of MWSs and fluency. In this model, speech production involves at least three different stages: *conceptualization*, *formulation*, and *articulation*. The conceptualization stage is where a preverbal message is generated. At this phase, the speaker's communicative intention is encoded into a coherent conceptual plan, while the message that is about to be sent is monitored. The preverbal message then moves to the formulation stage, where lexical selection and grammatical encoding take place. In this stage, the appropriate lemmas (i.e., lexical items unspecified for phonological form) are activated in the mental lexicon and placed into syntactic surface structures, and morphophonological and phonetic encoding are carried out. The product from these stages then moves to the articulation stage, where the phonetic plan is executed before speech is produced.

Compared to L1 processing and production, L2 processing and production are known to be challenged by the limitations of the L2 mental lexicon, which is "smaller, less organised, likely slower in access, less elaborated with syntactic and collocational information, and contains a narrower repertoire of formulaic language" (Skehan, Foster, & Shum, 2016, p. 98). A larger repertoire of formulaic language is therefore one way to reduce the load on the attentional resources that can be used in parallel processing at different stages of speech production (Kormos, 2006; Skehan, 2014).

More specifically, the benefit of knowledge and use of MWSs for oral fluency can be realized in lexical selection at the formulation stage (Kormos, 2006; Levelt, 1992). In lexical selection, speakers retrieve an appropriate lemma from a myriad of alternatives in the mental lexicon. At this stage, speakers with a large repertoire of MWSs can retrieve longer units of word constituents with similar processing cost to that needed for retrieval of single-word items. This would hypothetically allow them to "buy" processing time in preparation for other processing needs, such as syntactic processing and subsequent message generation (Boers et al., 2006; Skehan, 1998). In contrast, speakers with a smaller repertoire of MWSs may not enjoy such a processing advantage because they may exhaust cognitive resources in an attempt to retrieve every constituent item of the whole multiword unit. In addition, it can be hypothesized that phraseologically proficient speakers are less likely to show hesitations or pauses within MWSs because phrase structures are relatively fixed and less likely to be subjected to significant grammatical modification (Boers et al., 2006). Therefore, it can be argued that the fluid functioning of the formulator depends to a great extent on the size and organization of the mental lexicon and the mapping of lemmas to concepts. This is where MWSs are purported to enhance fluency because they facilitate access and retrieval of lexical units and free up attentional resources that are needed to deal with other aspects of speech performance.

Empirical Evidence for the Relationship

In the field of SLA research, there is a growing interest in investigating the relationship between the use of MWSs and L2 proficiency (Boers & Webb, 2018), which has been motivated by the strong theoretical and empirical psycholinguistic evidence discussed above. However, the majority of studies in this area have focused on written production using an essay-writing task (e.g., Bestgen & Granger, 2014; Durrant & Schmitt, 2009; Granger & Bestgen, 2014; Kim, Crossley, & Kyle, 2018; Kyle & Crossley, 2015; Kyle, Crossley, & Berger, 2018; Laufer & Waldman, 2011; Paquot, 2019; Siyanova-Chanturia & Spina, 2019). The findings of these studies suggest that L2 writers of higher proficiency, compared to those of lower proficiency, tend to have richer phraseological knowledge as well as better control of MWSs in communication.

Three recent studies conducted by Crossley and colleagues (Garner & Crossley, 2018; Kim et al., 2018; Kyle & Crossley, 2015) have shed light on the relationship between oral proficiency and MWSs. Adopting corpus-based measures to identify recurrent two-word and three-word sequences (bigrams and trigrams) in terms of proportion, frequency, and associational strength (e.g., *t* score, mutual information [MI] score), the researchers examined the extent to which L2 speakers employ target like MWSs. A cross-sectional study conducted by Kyle and Crossley (2015) found that holistic oral proficiency scores (human ratings) were significantly and positively correlated with a range of n-gram frequency and proportion scores, indicating that orally proficient speakers produce a greater number of target like sequences frequently used by L1 speakers. Two longitudinal studies conducted by Kim et al. (2018) and Garner and Crossley (2018) confirmed that L2 speakers produce a gradually increasing proportion of bigrams and trigrams frequently found in a L1 spoken reference corpus. The results also suggested that L2 speakers produce more high-frequency bigrams over time. However, L2 speakers showed little or no change in n-gram association scores (*t* score and MI) with the development of proficiency (see also Siyanova-Chanturia & Spina, 2019). This finding is in line with Bestgen and Granger's (2014) study highlighting a lack of longitudinal change in association scores in L2 writing. In sum, the existing literature supports the view that there is a positive relationship between general oral proficiency and the amount of use of MWSs, but not necessarily between proficiency and the type of MWSs used.

Some studies have more specifically examined the relationship between oral fluency and use of formulaic language (Boers et al., 2006; Stengers et al., 2011; Tavakoli, 2011; Wood, 2009, 2010). Overall, these studies have shown significant positive relationships between use of MWSs and oral fluency. Boers et al. (2006) elicited speech samples from 17 learners of English as a foreign language in Belgium through two speaking tasks (retelling and narrative), and had the samples rated for fluency by an experienced teacher. The study found a significant and moderate correlation between MWS counts and perceived fluency scores ($r = .393$, $p < .05$). Like Boers et al., Stengers et al. (2011) elicited speech samples through a retelling task from two groups of L2 learners in Belgium (26 studying English and 34 studying Spanish). The study found medium-to-large correlations between fluency rating scores and MWS counts for both learners of English ($r = .550$, $p < .01$) and learners of Spanish ($r = .361$, $p < .05$). As opposed to the subjective human rating approach adopted by Boers et al. and Stengers et al., Wood (2009, 2010) assessed fluency objectively using multiple temporal measures, including speech rate (i.e., mean number of syllables per minute) and mean length of run (i.e., total number of syllables or words per run in a sample, where "run" refers to a stretch of language between two pauses). Working with one Japanese ESL learner in Canada over the period of a 6-week fluency instruction course, Wood (2009) reported that fluency improvement took place in parallel with an increase in the number of MWSs. Wood (2010) reported similar findings when examining the development of MWSs among 11 ESL learners over a 6-month study in Canada. In line with Wood (2009), the study found a significant improvement in a range of speed fluency measures (e.g., speech rate, mean length of run), which was accompanied by a significant increase in the use of MWSs in speech.

Although the emerging research evidence summarized above highlights a positive relationship between oral fluency and MWSs, we find this evidence limited in a number of ways. First, some of these studies were based on relatively small sample sizes ($N = 1$ in Wood, 2009; $N = 11$ in Wood, 2010) or on participants with a restricted range of L2 proficiency levels (upper-intermediate to advanced levels in Boers et al., 2006, and B2 level of the Common European Framework of Reference in Stengers et al., 2011). These limitations in sample size and the restricted range of proficiency might have an impact on the generalizability of the findings. Second, the measurement of fluency in earlier studies (Boers et al., 2006; Stengers et al., 2011) was based on subjective judgments of fluency. Given the interest in the field of language testing in moving toward a more objective measurement of L2 ability (Tavakoli, Nakatsuhara, & Hunter, 2017), it is important to employ more objective measurements of fluency (Thomson, Boers, & Coxhead, 2017). The few studies that have examined the relationship between objective measures of utterance fluency and MWSs have assessed fluency in a rather limited way by examining only one aspect of fluency, for example, speed fluency (e.g., Wood, 2009).

The Current Study

In the current study we aimed to improve understanding of how the use of MWSs by L2 speakers relates to their oral fluency at different proficiency levels. The following research questions guided the study:

- To what extent does use of MWSs measured through n-grams (proportion, frequency, and strength of association) relate to level of proficiency in the TEEP speaking test?
- To what extent is use of MWSs measured through n-grams (proportion, frequency, and strength of association) associated with oral fluency (speed, breakdown, and repair) in the TEEP speaking test?

Method

TEEP Speaking Test

The TEEP is a standardized proficiency test used by a number of universities in the United Kingdom to obtain information about candidates' proficiency before they start a university degree. The TEEP speaking test includes three tasks, all on the same topic or question. The first task, used to elicit data in the current study, is an extended monologic task in which the test-taker is expected to speak for 3 minutes about a given topic. Unlike some other international tests providing a 30-second planning time (e.g., Aptis), the TEEP provides a longer planning opportunity before Part 1 and Part 2 tasks in the speaking test. This opportunity seems to reduce the cognitive load and communicative pressure of the task by allowing test-takers to plan what they want to say. Table 1 provides

Table 1 Structure of the speaking test of the Test of English for Educational Purposes

Part	Task	Mode	Example	Planning time	Response time
1	Individual talk (role plays)	Monologue	Question: Which is better: private or public services?	4 minutes	3 minutes
2	Scenario discussion	Dialogue	In pair, discuss with your partner and analyze the question.	2 minutes	4 minutes
3	Focus question	Further discussion	Discuss the question further with your partner, and agree or disagree!	None	No time limit but generally about 2 minutes

information about the three tasks within the TEEP speaking test. Unfortunately, for reasons of test security, we cannot provide task content or instructions.

The speaking section of the test is rated on a 9-point scale ranging from 0, where the speaker makes no attempt to speak, to level 8, where the speaker is very proficient. Levels 3 and below are labeled "*limited speaker*," and the highest level is labeled "*very good speaker*." Drawing on both holistic and analytic rating scales, the rating scales for the TEEP speaking test assess performance against the following criteria: *explaining ideas and information*, *interaction*, *fluency*, *accuracy and range*, and *intelligibility*. Test-takers' performance is examined by two[1] trained examiners, one acting as an interlocutor and the other as an assessor. The interlocutor introduces the questions and provides guidance before the conversation starts, but does not participate in the conversation. The assessor acts as an observer, sitting quietly at the back of the room listening and examining the test. Interlocutors provide holistic grades, namely, *explaining ideas and information* and *interaction*, whereas the assessors give both holistic grades for the same criteria and analytical grades for *fluency*, *accuracy and range*, and *intelligibility*. The assessors and interlocutors work with a set of validated marking scales and marking descriptors for each of the five criteria mentioned above to assess the candidate's performance. The holistic criteria (*explaining ideas and information* and *interaction*) are weighted more heavily than the analytic criteria (*fluency*, *accuracy and range*, and *intelligibility*). For further information about the test and to see samples of past papers, please visit https://www.reading.ac.uk/ISLI/study-in-the-uk/tests/isli-test-teep.aspx. All performances are recorded by the test administration team.

Data

The data for the current study were provided by the International Study and Language Institute at the University of Reading. The data came from 56 test-takers taking the TEEP Speaking paper (all three parts), which had been assessed by two (or at times three) experienced raters and placed on the 9-point rating scale. Based on the assessment of their speaking skills, the test-takers' spoken proficiency levels had been rated at four levels, namely, 5.0, 5.5, 6.5, and 7.5, which are equivalent to levels Low B1, High B1, B2, and C1, respectively, of the Common European Framework of Reference (Council of Europe, 2001). While these proficiency level scores were based on the overall assessment of the test-takers' performance on the TEEP speaking test, the data used for measuring utterance fluency came from the test-takers' performance of just Task 1 (i.e., an extended monologic task). This data set comprised 56 task performances, totaling 168 min of recordings (i.e., 56 performances $\times$ 3 minutes). There were 11 participants at Low B1, 14 at High B1, 17 at B2, and 14 at C1 level of proficiency. The test-takers were pre-degree university applicants who took the test as one of the entry requirements to their programs of study at a British university. The test-takers came from a range of 10 different L1 backgrounds, including Chinese, Arabic, Japanese, Kazakh, Thai, Greek, Turkish, and Portuguese.

Measuring Fluency

Following recent fluency research, we measured utterance fluency in terms of speed, breakdown, and repair fluency (Kahng, 2014; Skehan, 2003; Tavakoli & Skehan, 2005). Different studies have shown that these three aspects are distinct factors underlying the fluency construct (Kahng, 2014; Skehan & Foster, 1997; Tavakoli & Skehan, 2005). For each aspect of fluency, there are several measures that can be used. That is, there are different measures to reflect the speaker's degree of speed (e.g., speech rate, articulation rate, and phonation time ratio), breakdown (e.g., length and frequency of filled and silent pauses), and repair (e.g., frequency of disfluency markers, repetitions, and self-corrections). However, researchers have been warned against using several measures from each aspect because they may correlate and overlap with each other (Bosker, Pinget, Quené, Sanders, & de Jong, 2013; Kahng, 2014; Skehan, 2009). To avoid this problem, we followed the recent research findings in this area to choose our fluency measures.

We selected articulation rate, that is, total number of syllables divided by total amount of speaking time per minute (excluding pauses), as a reliable representation of speed fluency (de Jong, Groenhout, Schoonen, & Hulstijn, 2015; Kahng, 2014; Mora & Valls-Ferrer, 2012; Suzuki & Kormos, 2019). We chose frequency of silent pauses per minute to examine breakdown fluency, following recent research in this area (de Jong et al., 2015; Kahng, 2014; Skehan, 2014; Suzuki & Kormos, 2019). We also examined pause location in terms of occurrence in the middle of the clause or at the end of the clause. Research in this area (Skehan et al., 2016; Tavakoli, 2011; Tavakoli, Nakatsuhara, & Hunter, 2020) suggests that L2 speakers, especially at lower proficiency levels, pause more frequently in mid-clause position, whereas native speakers pause more frequently at end-clause junctures. Some emerging evidence also suggests that there is a link between lexical knowledge and pause locations (de Jong, 2016): That is, learners with limited L2 lexical knowledge are more likely to pause in both mid-clause and end-clause positions. Following de Jong and Bosker (2013), a pause in the current study is defined as an instance of silence longer than 250 milliseconds. Finally, following Hunter (2017) and Skehan (2009), we used the total number of repairs per minute to reflect repair fluency, with repairs comprising hesitations, repetitions, reformulations, and self-corrections.

Measuring MWSs

There are generally two approaches to defining and measuring MWSs: a phraseological and a frequency-based approach (Boers & Webb, 2018; Granger & Paquot, 2008). A phraseological approach often involves judgments of formulaicity according to a range of identification criteria, including semantic opacity of individual words comprising MWSs (e.g., *kick the bucket*), phonological structure of word strings (e.g., vowel reduction, assimilation), and/or grammatical irregularity (e.g., *if I were you*; Howarth, 1998; Myles & Cordier, 2017; Wood, 2009, 2010; Wray, 2002). A major issue with this approach is that it involves a fair degree of subjectivity, which might impact on the reliability of the analysis. Notably, earlier studies showed relatively low agreement between trained judges on judging units of MWSs ($r < .60$ in Boers et al., 2006, and Stengers et al., 2011), falling far short of the median interrater reliability in SLA research ($r = .92$; Plonsky & Derrick, 2016).

In contrast, a frequency-based approach determines formulaicity of word combinations according to the frequency of two or more words co-occurring in an external reference corpus (Gablasova, Brezina, & McEnery, 2017). For instance, the corpus-based approach identifies consecutive and recurrent sequences of a given number of words (i.e., n-grams), including grammatically complete or incomplete word combinations, such as bigrams (e.g., *in the*, *think that*) and trigrams (e.g., *one of the*, *the fact that*). Because of the relatively large sample size in the current study ($N = 56$) compared to previous fluency studies ($N = 1$ to 34 in Boers et al., 2006; Stengers et al., 2011; Wood, 2009, 2010) and the potential reliability issue for human judgments of formulaicity, we adopted a frequency-based measure of recurrent word combinations or n-grams in order to identify MWSs objectively in our data.

N-Gram Analysis

Following earlier phraseological research (Garner & Crossley, 2018), we employed three n-gram indices—proportion, frequency, and association—based on two- and three-word contiguous chunks (i.e., bigram and trigram tokens). To calculate n-gram scores, we used the Tool for the Automatic Analysis of Lexical Sophistication 2.0 (TAALES: Kyle & Crossley, 2015; Kyle et al., 2018). TAALES is an up-to-date computational tool used to assess various facets of learner performance in terms of lexical sophistication, such as word frequency, range, and psycholinguistic word information. We selected the spoken subsection of the Corpus of Contemporary American English (COCA; Davies, 2009) as a reference corpus with which to calculate a set of n-gram indices. Our selection of COCA was based on the relatively large size of its spoken subsection, comprised of 79 million words from spontaneous conversations from a wide range of TV and radio programs in the United States recorded over the last 25 years.

N-Gram Measures

As mentioned above, we used TAALES to calculate three types of n-gram measures (proportion, frequency, and association), producing a total of eight score indices (two proportion, two frequency, and four association indices). Instead of setting a cut-off point to divide lexical combinations into MWSs and non-MWSs, we computed mean frequency and association scores. We took the view that MWSs should be described on a grading scale of frequency and associative strength rather than by any dichotomous classification of formulaicity (e.g., frequent vs. infrequent, associated vs. not associated; Durrant & Schmitt, 2009; Ellis, 2012).

N-Gram Proportion Measures

Proportion score indices refer to the proportion of bigrams and trigrams in learner spoken data that are also found in the representative corpora. In this study, TAALES calculated the proportion of bigrams and trigrams produced by the test-takers in their speech sample that were among the 30,000 most frequent n-grams in the spoken subsection of COCA. Higher proportion scores indicate that speakers produce a higher concentration of target like, high-frequency bigrams and trigrams when text length is controlled for. Previous research suggests that high-proficiency L2 users show a greater proportion of target-like n-grams in their language production (Garner & Crossley, 2018; Kyle & Crossley, 2015).

N-Gram Frequency Measures

Unlike n-gram proportion indices, which are based on binary scoring (i.e., presence or absence of learner-produced n-grams in the reference corpora), frequency score indices base their scoring on a continuous scale. Based on frequency in a referenced corpus, they assign a frequency score to all the bigrams and trigrams in a learner text and average all assigned scores to yield a single composite score per speaker. Following Kyle and Crossley's (2015) suggestion, we used logarithmic bigram and trigram frequency scores instead of raw frequency scores in order to control for Zipfian effects common in word frequency lists. Higher frequency scores indicate that speakers produce a larger number of high-frequency, target like n-grams when text length is controlled for. Whereas earlier research in this area has suggested that more proficient L2 users produce a greater number of high-frequency n-grams (Garner & Crossley, 2018; Kyle & Crossley, 2015) than less proficient speakers, Siyanova-Chanturia and Spina's (2019) study implies that high-proficiency learners rely to a greater extent than less proficient learners on low-frequency MWSs in which constituent words are non-associated.

N-Gram Association Measures

N-gram association score indices show the strength of association between individual words comprising bigrams or trigrams. Association indices are similar to proportion and frequency indices in that they are all based on frequencies of word co-occurrence found in a given corpus. However, association measures are less strongly influenced by the individual frequencies of constituent words because those frequencies are controlled for in calculating association scores.

Of five association measures available in TAALES, we selected *t* score and MI, two measures that have been extensively used in SLA research investigating MWSs in both spoken and written production (e.g., Durrant & Schmitt, 2009; Garner & Crossley, 2018; Granger & Bestgen, 2014; Saito, 2019). These two measures are similar in comparing the observed frequency of n-grams in the reference corpus (COCA in this study) with the expected frequency computed on the basis of the frequency of the constituent words (Evert, 2005). However, the two measures tend to yield insight into different sets of word combinations. MI highlights combinations "which may be less common, but whose component words are not often found apart" (e.g., *ultimate arbiter*, *tectonic plates*), whereas *t* score highlights very frequent combinations, whose rankings "are very similar to rankings based on raw frequency" (e.g., *good example*, *hard work*; Durrant & Schmitt, 2009, p. 167). Research has suggested that L2 essays written by high-proficiency writers contain word combinations receiving higher MI scores (MWSs of greater sophistication) but lower *t* scores (fewer easy MWSs) than essays written by low-proficiency writers (e.g., Bestgen & Granger, 2014; Durrant & Schmitt, 2009; Granger & Bestgen, 2014). However, no research has so far shown any significant relationship between these association measures and oral proficiency.

Data Analysis

The first step in the data analysis was to transcribe the spoken data before coding them for fluency and n-gram measures (see discussion above for details of the measures). The unpruned transcriptions were segmented into AS-units (analysis of speech units; Foster, Tonkyn, & Wigglesworth, 2000) and clauses. This enabled us to investigate pauses in mid- and end-clause positions. For the analysis of fluency, PRAAT (Boersma & Weenink, 2013) was used to identify the measures of speed and pausing. The data were annotated manually on PRAAT by an expert researcher with extensive experience of analyzing similar data. The annotation process involved several stages, including listening to the extracts of speech, inspecting the spectrograms produced in PRAAT, and identifying and tagging fluency features (e.g., pauses and repairs) on a corresponding grid.

Although PRAAT measurement is precise and accurate, the entire data set was subjected to PRAAT analysis for a second time to ensure a high level of intrarater reliability. The repair measures were also coded manually by one of the researchers, and 20% of the transcripts were coded for a second time by another researcher to check the interrater reliability. A reliability coefficient of .95 was obtained between the first and second coding for the repair measures. Prior to n-gram analysis using TAALES (Kyle & Crossley, 2015), all the transcripts used for fluency analysis were inspected to fix obvious pronunciation errors and remove orthographic markings of filled pausing (e.g., *uh*, *um*), which is a prerequisite for lexical analysis. The resulting transcripts ranged between 93 and 403 words in length ($M = 260.8$, $SD = 72.2$).

Results

Before running correlation analyses to answer the research questions, we examined the descriptive statistics of the data set. Table 2 provides the means, standard deviations, medians, minimums, and maximums for the different fluency and n-gram measures across proficiency levels. The figures in this table suggest that as proficiency develops, the speed of performance increases and the frequency of pauses, at both mid- and end-clause locations, decreases. The picture is less clear regarding total repair, which does not show a consistent pattern. As for n-gram units of analysis, the descriptive statistics suggest that frequency, proportion, and the association measure of *t* score increase as proficiency develops. However, the association measure of MI generally decreases with the development of proficiency. We discuss these observations further in the following sections.

Use of MWSs at Different Proficiency Levels

Our first research question was aimed at examining the use of MWSs at different proficiency levels. To examine the relationship between oral proficiency and n-gram measures, we conducted a Kruskal–Wallis *H* test. We adopted this non-parametric test in response to the violation of homogeneity of variance across the four proficiency groups (Levene statistic = 4.43 to 17.84, $p = .008$ for bigram frequency, $p = .001$ for bigram *t*, and $p < .0001$ for other measures). The analysis showed that there were significant differences for six n-gram measures across the proficiency levels: bigram frequency, $\chi^2(3) = 48.13$, $p = .001$; bigram MI, $\chi^2(3) = 15.50$, $p = .001$; bigram *t*, $\chi^2(3) = 12.54$, $p = .006$; trigram frequency, $\chi^2(3) = 28.45$, $p = .001$; trigram *t*, $\chi^2(3) = 31.18$, $p = .001$; trigram proportion, $\chi^2(3) = 20.36$, $p = .001$. There were no significant differences for the remaining two n-gram measures: bigram proportion, $\chi^2(3) = 4.95$, $p = .175$, and trigram MI, $\chi^2(3) = 7.15$, $p = .067$.

Post-hoc tests (Mann–Whitney *U* tests) with Bonferroni correction (0.05/6 for six comparisons per measure) revealed the following patterns at the $p < .008$ level: bigram frequency (Low B1 = High B1 < B2 < C1); bigram MI (High B1 = B2 > C1); bigram *t* (B2 < C1); trigram frequency (Low B1 < B2 < C1); trigram *t* (Low B1 < B2 < C1); and trigram proportion (Low B1 < B2 < C1). The analysis suggested that for most measures, there was likely a linear relationship between general oral proficiency level and n-gram measures. Figures 1–8 provide visual representations of the relationship between speaking proficiency and n-gram measures of frequency, proportion, and association (see Appendix S1 in the Supporting Information online for effect sizes, confidence intervals, and *p* values). Appendix S2 provides exact

Table 2 Descriptive statistics for all fluency and n-gram measures at different levels of proficiency ($N = 56$)

Measure	Level[a]	*M* (*SD*)	Median	Minimum	Maximum
Articulation rate	LB1	216.75 (24.67)	215.14	187.703	270.06
	HB1	224.99 (17.21)	224.25	199.29	257.34
	B2	231.77 (17.80)	230.24	203.85	260.59
	C1	240.11 (25.10)	239.39	193.50	285.29
Frequency of mid-clause pauses	LB1	12.52 (3.19)	14.40	6.07	15.69
	HB1	13.54 (2.61)	14.35	7.35	16.73
	B2	11.37 (3.55)	10.89	5.96	18.11
	C1	8.94 (3.66)	9.00	3.69	13.16
Frequency of end-clause pauses	LB1	24.56 (4.27)	24.75	17.95	30.75
	HB1	24.38 (3.77)	23.59	17.92	30.87
	B2	19.52 (4.18)	19.34	12.89	29.20
	C1	19.65 (4.24)	18.83	12.18	27.46
Frequency of total repair	LB1	5.91 (3.30)	4.88	1.21	10.40
	HB1	4.63 (5.04)	3.01	0.00	17.06
	B2	5.72 (2.49)	4.73	1.67	9.82
	C1	5.19 (2.87)	5.40	1.14	10.35
Bigram log frequency	LB1	1.48 (0.10)	1.48	1.30	1.61
	HB1	1.54 (0.06)	1.57	1.42	1.62
	B2	1.67 (0.03)	1.68	1.62	1.73
	C1	1.78 (0.03)	1.79	1.74	1.83
Bigram proportion	LB1	0.50 (0.07)	0.51	0.34	0.58
	HB1	0.52 (0.06)	0.55	0.40	0.60
	B2	0.52 (0.02)	0.52	0.46	0.55
	C1	0.53 (0.00)	0.53	0.53	0.54
Bigram association MI	LB1	1.47 (0.20)	1.55	1.11	1.69
	HB1	1.47 (0.15)	1.38	1.29	1.78
	B2	1.38 (0.05)	1.39	1.17	1.41
	C1	1.36 (0.01)	1.36	1.34	1.38
Bigram association *t*	LB1	55.34 (25.57)	58.39	−7.76	83.93
	HB1	62.21 (19.86)	59.13	14.32	95.57
	B2	62.41 (9.58)	65.01	26.61	67.90
	C1	70.62 (1.51)	70.62	68.27	72.97
Trigram log frequency	LB1	0.83 (0.13)	0.85	0.58	0.98
	HB1	0.91 (0.17)	0.84	0.69	1.21
	B2	0.96 (0.02)	0.96	0.93	0.99
	C1	1.02 (0.02)	1.02	0.99	1.05

(*Continued*)

Table 2 Continued

Measure	Level[a]	*M* (*SD*)	Median	Minimum	Maximum
Trigram proportion	LB1	0.13 (0.05)	0.13	0.05	0.22
	HB1	0.16 (0.05)	0.17	0.09	0.25
	B2	0.17 (0.01)	0.17	0.16	0.18
	C1	0.19 (0.01)	0.19	0.18	0.19
Trigram association MI	LB1	2.24 (0.31)	2.22	1.86	2.76
	HB1	2.27 (0.25)	2.31	1.87	2.82
	B2	1.38 (0.05)	2.21	1.83	2.22
	C1	2.19 (0.01)	2.19	2.18	2.20
Trigram association *t*	LB1	19.59 (7.88)	20.53	7.88	30.79
	HB1	30.41 (12.96)	26.55	18.12	59.32
	B2	30.61 (4.53)	31.50	13.92	33.96
	C1	36.26 (1.29)	36.26	34.26	38.26

[a]Proficiency level in the Common European Framework of Reference. LB1 = Low B1; HB1 = High B1. I = mutual information.

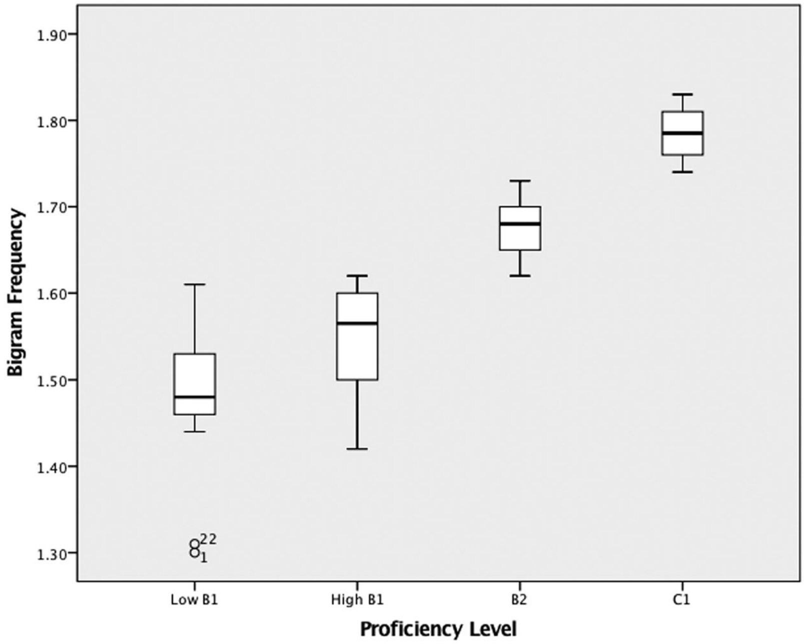

Figure 1 Bigram frequency plotted against proficiency level of the Common European Framework of Reference.

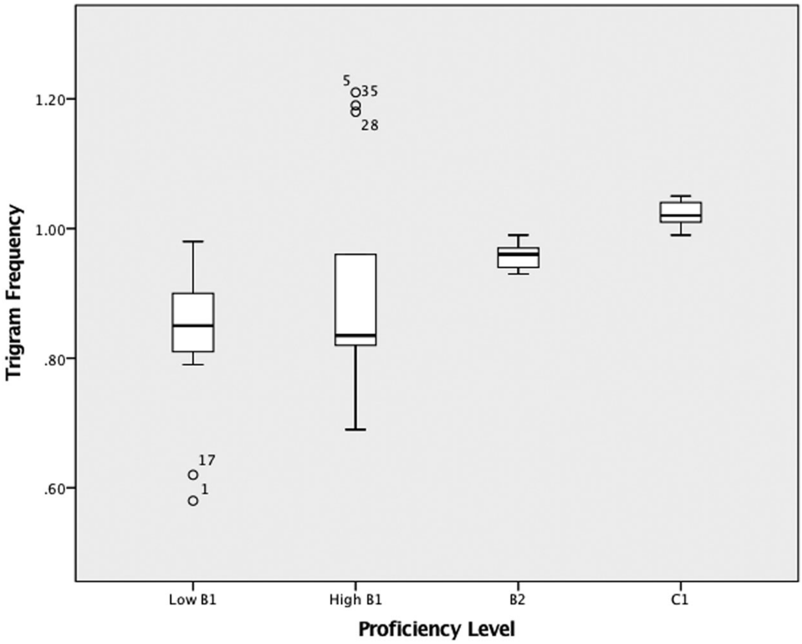

Figure 2 Trigram frequency plotted against proficiency level of the Common European Framework of Reference.

p-values for each comparison of the n-gram measures across the four proficiency levels.

Relationship Between Fluency and N-Gram Units

Our second research question asked whether there was a relationship between different aspects of fluency (speed, breakdown, and repair) and different n-gram measures (proportion, frequency, and association). In total, we had four fluency measures and eight n-gram measures. Having based our selection of fluency measures on the findings of previous research (Kahng, 2014; Suzuki & Kormos, 2019), we believe these measures are valid representatives of the three aspects of fluency. However, we were concerned that running multiple correlations between each of these fluency measures and all of the n-gram measures would run a risk of making a Type I error, given the small sample size of the study ($N = 56$). Thus, in order to answer the second research question, we first ran a principal component analysis to see how many factors might underlie the eight

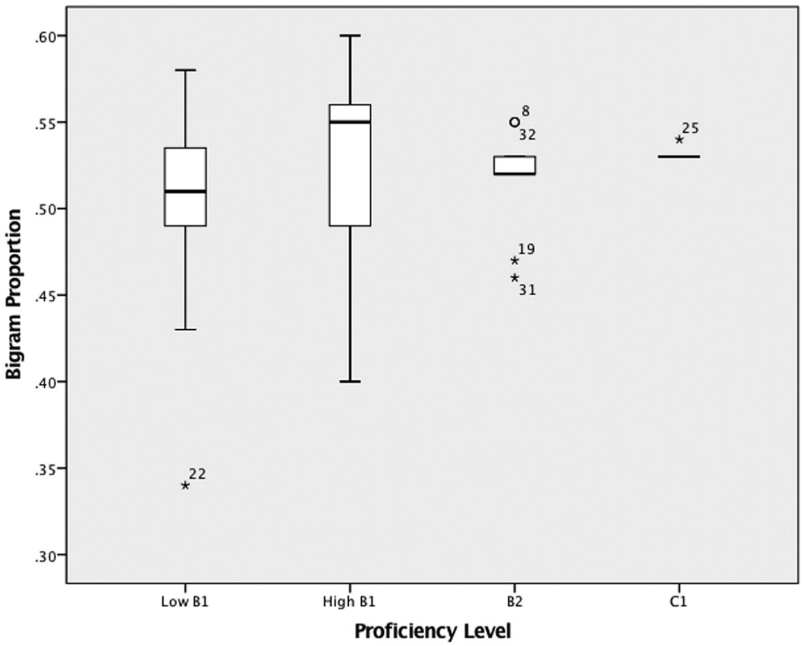

Figure 3 Bigram proportion plotted against proficiency level of the Common European Framework of Reference.

n-gram measures. We discuss the results of this principal component analysis before reporting the correlation analyses.

Data Reduction: Underlying Factors of N-Gram Units

To control for the potential overlap between the MWS measures, we submitted all eight n-gram measures to a principal component analysis with varimax rotation. The first attempt of the principal component analysis produced four factors, one of which was explained by a single n-gram measure alone (namely, bigram *t* score). In order to avoid the issue of factor underdetermination (i.e., factors not being represented by multiple measured variables; Fabrigar, Wegener, MacCallum, & Strahan, 1999), we conducted an additional principal component analysis using all n-gram measures excluding the bigram *t* score. The factorability of the entire dataset (except the bigram t score) was confirmed using two tests: Bartlett's test of sphericity ($\chi^2 = 230.89$, $p = .001$) and the Kaiser–Meyer–Olkin measure of sampling adequacy (.647). Based on eigenvalues beyond 0.7 (Stevens, 2002) and the visual inspection of the scree

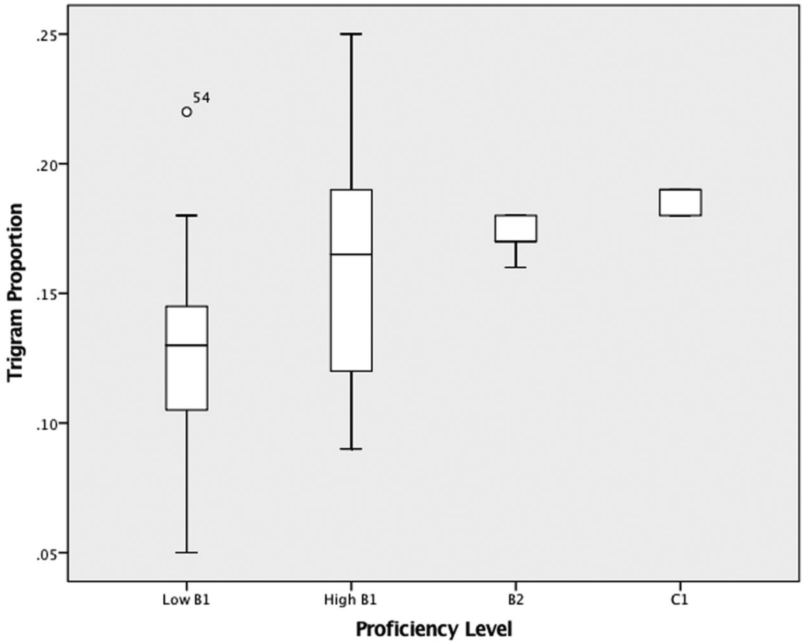

Figure 4 Trigram proportion plotted against proficiency level of the Common European Framework of Reference.

plot, three factors were identified in the data set, accounting for 86.1% of the variance in the n-gram measures (without the bigram *t* score) (summarized in Table 3).

We labeled Factor 1 "High-Frequency Trigram" because it encompassed trigram measures that characterize high-frequency combinations of words, that is, trigram frequency and association measure of *t*, which has been reported to highlight high-frequency combinations (Durrant & Schmitt, 2009). Although the result showed moderate sizes of loadings of bigram frequency ($r = .604$) and trigram proportion ($r = .616$) on Factor 1, we labeled this factor according to the two trigram measures that had higher loadings ($r > .90$).[2] Factor 2 was labeled as "Association Measure of MI" because the result showed positive loadings of the two association measures (bigram and trigram MI scores) on this factor. The negative loading of bigram log frequency on this factor also looks reasonable given that MI scores have been reported to highlight the low-frequency nature of the n-grams (Gablasova et al., 2017). Factor 3 was labeled "Proportion" because it captured bigram and trigram proportion scores. Given

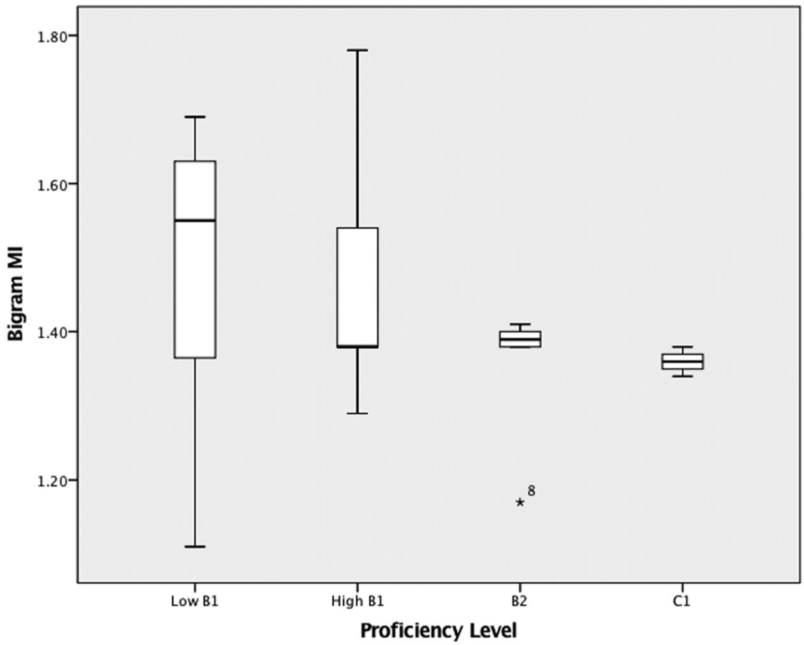

Figure 5 Bigram MI (mutual information) plotted against proficiency level of the Common European Framework of Reference.

the small sample size of the study, we suggest that the results of this factor analysis should be considered cautiously. Table 3 shows all the loadings for the three underlying factors.

Correlations Between Fluency and N-Gram Factors

To examine the relationship between fluency and n-gram units, we conducted a series of simple Pearson correlation analyses with a Bonferroni correction of $\alpha = .017$ (from 0.05/3 for three comparisons per measure) between the resulting three n-gram factors and fluency measures of speed, breakdown, and repair (Table 4). It is plausible to argue that unpruned data containing several repair features may not accurately represent speed fluency. To address this concern, we decided to include both pruned and unpruned measures of articulation rate to examine their relationship with the n-gram factors. To ensure there were no violations of the assumptions of normality, linearity, and homoscedasticity, we performed preliminary analyses by visual inspection of histograms, scatterplots, P–P plots, and plots of studentized residuals crossed

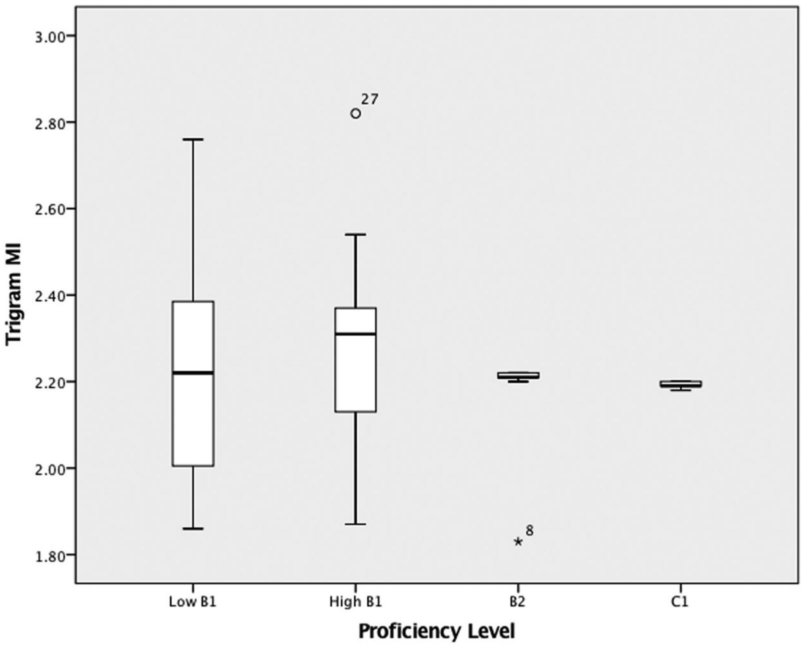

Figure 6 Trigram MI (mutual information) plotted against proficiency level of the Common European Framework of Reference.

with fitted values (see Larson-Hall, 2010, p. 160). In line with Plonsky and Oswald's (2014) recommendation, we interpret correlation coefficients of $0.25 \le r < 0.40$ as small, $0.40 \le r < 0.60$ as medium, and ≥ 0.60 as large.

Significant but small to medium strength correlations were observed between High-Frequency Trigram and pruned articulation rate ($r = .396$, $p = .003$) and unpruned articulation rate ($r = .325$, $p = .014$), suggesting that more fluent speakers produced more high-frequency n-grams. For frequency of mid-clause pauses, there was a negative correlation with the n-gram Proportion factor ($r = -.271$, $p = .041$), but given our corrected alpha level of .017 the correlation is not considered statistically significant, although a small effect size of 0.271 (Plonsky & Oswald, 2014) was observed. Similarly, the positive correlation between the number of end-clause pauses and the Association Measure of MI ($r = .302$, $p = .024$), and the negative correlation between MI and frequency of total repair ($r = -.308$, $p = .021$) miss reaching the corrected alpha level. In terms of the magnitude of effect sizes, these results indicate small associations ($0.25 \le r < 0.40$) and could tentatively imply that those

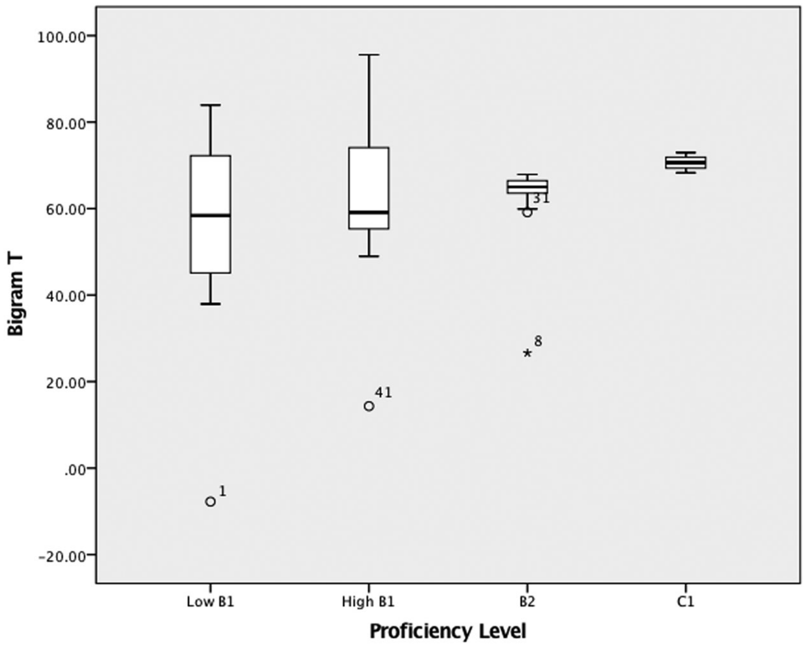

Figure 7 Bigram T plotted against proficiency level of the Common European Framework of Reference.

speakers who produced n-grams of high mutual information quality tended to pause slightly more frequently at end-clause pauses and made fewer repairs to their utterances.

Qualitative Analysis of MWSs Used Across Different Proficiency Levels

To gain a deeper understanding of the role MWSs play in the development of oral proficiency, we examined the profiles of the most frequently used bigrams and trigrams across proficiency levels (Low B1, High B1, B2, and C1). Tables 5 and 6 show the 20 most frequent bigrams and 20 most frequent trigrams, respectively, used by the group of participants as a whole ($N = 56$). See Appendices S3-S6 in the Supporting Information online for breakdowns of the most frequent MWSs for the proficiency groups: Low B1 ($n = 11$), High B1 ($n = 14$), B2 ($n = 17$), and C1 ($n = 14$) groups, respectively.

As indicated in the tables, in general all the test-takers, regardless of their proficiency level, seem to have used similar types of bigrams and trigrams, including both grammatically complete sequences (e.g., *historical sites*) and

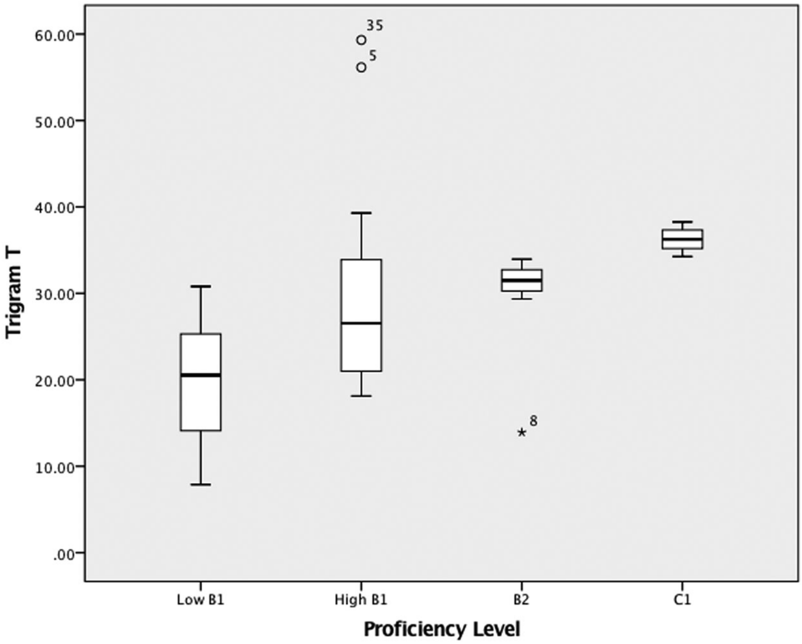

Figure 8 Trigram T plotted against proficiency level of the Common European Framework of Reference.

Table 3 Principal component analysis of the n-gram measures

Measure	Factor 1 (High-Frequency Trigram)	Factor 2 (Association Measure of MI)	Factor 3 (Proportion)
Trigram log frequency	**.920**	−.089	.223
Trigram *t* score	**.910**	.198	.125
Bigram MI	−.082	**.894**	.038
Trigram MI	.404	**.764**	−.278
Bigram log frequency	**.604**	**−.614**	.225
Bigram proportion (30,000)	.130	−.125	**.947**
Trigram proportion (30,000)	**.616**	−.062	**.711**

Note. Factor loadings >.600 are shown in boldface. MI = mutual information.
The proportion of bigrams/trigrams in a sample text that are among the 30,000 most frequent trigrams in COCA (Corpus of Contemporary American English).

Table 4 Pearson correlations between fluency measures and n-gram factors

	N-gram factor					
	Factor 1		Factor 2		Factor 3	
	(High-Frequency Trigram)		(Association Measure of MI)		(Proportion)	
Fluency measure	*r*	*p*	*r*	*p*	*r*	*p*
Articulation rate (unpruned)	.325	.014	−.031	.821	.136	.316
Articulation rate (pruned)	.396	.003	−.067	.622	.165	.224
Frequency of mid-clause pauses	−.239	.076	.215	.112	−.271	.041
Frequency of end-clause pauses	−.202	.136	.302	.024	−.016	.905
Frequency of total repair	−.073	.593	−.308	.021	−.065	.633

Note. All *p*s are two-tailed. MI = mutual information.

grammatically incomplete sequences (e.g., *in the*, *and I think*). A closer inspection of the data, however, reveals some patterns emerging from the analysis. First, all the test-takers tended to use MWSs that were specifically relevant to task completion. Many of these MWSs—for example, *preserving historical sites*, *the historical sites*, and *sites for leisure*—seem to have been borrowed from the instructions used to guide the test-takers on what to discuss to complete the task. This observation is based on the test-takers' task performances (e.g., "As we know, our topic is about X").

As regards the frequent use of MWSs from task prompts, it should be noted that the novice speakers at a Low B1 level also used a large number of task-specific bigrams and trigrams (e.g., *historical sites*, *building sites for*, and *sites for leisure*). Despite this seemingly similar pattern in beginning (Low B1) and advanced (C1) learners regarding the use of task-related phrases, a closer inspection of the data indicated a qualitative difference between the two groups in the use of such MWSs. Whereas the Low B1 group borrowed the n-grams and used them repeatedly in their original form (e.g., *preserving historical sites*), the C1 level group borrowed the same n-grams but used them creatively by replacing some of the words with their synonyms (e.g., *protecting historical sites*), or by changing the structure of the n-grams (e.g., *to preserve these historical sites*). Overall, the speech of the lower proficiency speakers was typically characterized by repetition and redundancy of the MWSs, whereas C1-level speakers avoided repetition and recycled the MWSs competently.

Table 5 The 20 most frequent bigrams used by participants ($N = 56$)

		Text-internal data		Text-external (COCA) data		
Rank[a]	Bigram	Frequency	Range[b] ($N = 56$)	Frequency[c]	MI	t
1	I think	138	39	2,601.96	3.59	477.47
2	historical sites	104	32	—	—	—
3	in the	74	34	4,010.06	1.70	498.03
4	for the	73	34	1,248.01	1.28	245.58
5	and the	56	27	1,593.61	0.30	100.44
6	and it	54	29	801.70	0.77	146.11
7	of the	52	27	4,667.36	1.57	521.34
8	it can	51	24	73.80	0.72	42.34
9	preserving historical	51	30	—	—	—
10	good for	50	19	44.13	1.39	48.08
11	it is	50	23	856.32	1.57	222.97
12	they can	50	23	186.17	2.34	118.67
13	the country	48	26	308.93	2.20	150.71
14	for leisure	47	20	—	—	—
15	think it	47	27	440.90	2.02	175.41
16	lot of	46	17	916.72	3.56	283.18
17	a lot	45	16	1,156.60	3.79	319.99
18	for example	44	26	126.97	4.40	107.19
19	important for	41	27	21.08	1.93	37.81
20	sites for	40	18	—	—	—

Note. Contracted forms (e.g., *it's*) were counted as a single word; for example, *it's good* was considered to be a bigram rather than a quadgram. A dash indicates that the n-gram was not frequent enough in the reference corpus for a score to be calculated. MI = mutual information. [a]Frequency rank. [b]Range refers to the number of participants using the bigrams. [c]Frequency per 79 million words of the spoken subsection of COCA (Corpus of Contemporary American English).

Finally, the qualitative inspection of the data highlighted an interesting pattern with respect to n-gram use across different proficiency levels. The analysis revealed that the participants tended to use discourse markers (e.g., *for example*, *first of all, you know*) more frequently as proficiency increased. Table 7 provides token counts for the three most frequent discourse markers used by the participants, showing increases in the number of these discourse markers as proficiency increases.

Table 6 The 20 most frequent trigrams used by participants ($N = 56$)

		Text-internal data		Text-external data (COCA)		
Rank[a]	Trigram	Frequency	Range[b] ($N = 56$)	Frequency[c]	MI	*t*
1	I think it	46	26	314.05	3.68	166.31
2	a lot of	41	14	880.04	3.81	279.23
3	preserving historical sites	39	23	—	—	—
4	sites for leisure	35	16	—	—	—
5	building sites for	28	13	—	—	—
6	I think the	22	11	232.46	3.75	143.34
7	good for the	21	12	10.22	1.73	25.32
8	the historical sites	21	8	—	—	—
9	is good for	19	10	4.18	2.03	17.09
10	historical sites is	18	10	—	—	—
11	first of all	16	13	113.71	5.90	102.38
12	for national identity	16	15	—	—	—
13	important for national	16	15	—	—	—
14	so I think	15	12	87.73	2.06	78.70
15	in the past	14	7	75.09	3.43	80.71
16	of the country	13	10	52.87	2.08	61.29
17	it's good for	13	6	—	—	—
18	site for leisure	12	8	—	—	—
19	to talk about	12	11	104.93	2.54	90.81
20	very important for	12	8	4.82	4.53	20.92

Note. Contracted forms (e.g., *it's*) were counted as a single word; for example, *it's good for* was considered to be a trigram rather than a quadgram. A dash indicates that the n-gram was not frequent enough in the reference corpus for a score to be calculated. MI = mutual information. [a]Frequency rank. [b]Range refers to the number of participants using the trigrams. [c]Frequency per 79 million words of the spoken subsection of COCA (Corpus of Contemporary American English).

Discussion

In answer to the first research question, regarding the relationship between use of MWSs and general oral proficiency, we found evidence to suggest a potentially linear relationship between many of the n-gram measures and proficiency level. This result suggests that speakers at higher proficiency levels produced (a) a greater proportion of frequent n-grams, (b) more high-frequency n-grams, and (c) n-grams with lower MI scores. This finding is generally in line with previous

Table 7 Token counts of three most frequently used discourse markers across four oral proficiency levels

	Discourse marker		
Level[a]	*for example*	*first of all*	*you know*
Low B1 ($n = 11$)	1 (9%)	2 (18%)	2 (18%)
High B1 ($n = 14$)	6 (43%)	2 (14%)	2 (14%)
B2 ($n = 17$)	7 (41%)	4 (24%)	5 (29%)
C1 ($n = 14$)	11 (79%)	6 (43%)	7 (50%)

Note. Percentages indicate the tokens per number of speakers in the group.
[a]Proficiency level of the Common European Framework of Reference.

research (Garner & Crossley, 2018; Kim et al., 2018; Kyle & Crossley, 2015), but provides important additional evidence in support of the key role that MWSs play in oral proficiency. Given the scarcity of studies exploring the use of n-grams in oral performance across a range of proficiency levels, and the use in our study of a standardized speaking test to assess proficiency, the results significantly expand empirical knowledge in this area of SLA.

The results confirm previous findings that higher proficiency learners produce a greater proportion of n-grams and a greater number of high-frequency n-grams (Garner & Crossley, 2018; Kim et al., 2018; Kyle & Crossley, 2015). These findings support the view that language proficiency is closely linked with the quality of MWSs (i.e., high-frequency n-grams produced) as well as the amount of MWSs as a proportion of total productions (i.e., the number of n-grams produced). If proficiency develops through exposure and practice, learners who have more exposure and practice might become more sensitive to distributional patterns of language use (Ellis, Simpson-Vlach, & Maynard, 2008). This line of reasoning might explain why higher proficiency learners tend to use a larger number of MWSs that are also frequent in the speech of L1 users compared to lower proficiency learners.

However, the negative relationship between bigram MI scores and oral proficiency observed in this study was not in line with previous research in either writing or speaking studies. The writing studies, other than Siyanova-Chanturia and Spina (2019), have documented that higher proficiency learners produce more strongly associated n-grams indexed by higher MI scores (Durrant & Schmitt, 2009; Granger & Bestgen, 2014), whereas speaking studies have reported little to no meaningful relationship between MI scores and oral proficiency (Garner & Crossley, 2018; Kim et al., 2018). In the current study, the results suggested that the highest proficiency group, compared to lower levels of proficiency, did not produce a greater number of strongly associated n-grams (indexed by high MI scores). A possible reason for this unexpected finding might be linked to L2 beginners' repeated use of phrases with relatively high MI scores (e.g., MI ≧ 3; Hunston, 2002). A careful investigation of the transcripts of the data suggested that some of the n-grams—for example, *I think* (MI = 3.59)—were overused by learners of lower proficiency levels: there were 3.27 occurrences of the phrase per speaker for Low B1, 2.85 for High B1, 1.94 for B2, and 2.07 for C1.

Furthermore, learners' production of strongly associated n-grams appeared to be greatly affected by their use of phrases borrowed from task prompts. The frequent use of task-specific phrases was particularly obvious in C1 speakers' use of trigrams (e.g., *preserving historical sites*), the majority of which occurred too infrequently (≦5 occurrences) in the COCA spoken subcorpus for a score to be calculated (such atypical n-grams are marked with dashes in Tables 5 and 6). This meant that quite a few of the n-grams produced by C1 speakers (e.g., *the historical sites*, *preserving historical sites*, *sites for leisure*, *building sites for*, *historical sites is*) did not occur frequently enough in a reference corpus to receive either frequency or association scores computed by TAALES. Therefore, these n-grams were not considered in calculating the means of frequency or association scores per speaker. In fact, the correlation between frequency of n-grams used by C1 speakers and assigned MI scores is negligible (bigrams: r = .19; trigrams: r = .07), indicating that C1 speakers did not necessarily produce n-grams with higher MI scores repeatedly.

By contrast, the correlation in the groups of lower proficiency speakers is moderate, as is observed for Low B1 (bigrams: r = .39) and High B1 (bigrams: r = .43; trigrams: r = .47). The negligible correlation for Low B1 trigrams (r = .07) can similarly be explained in terms of the participants' heavy reliance on task-specific key-phrases (e.g., *building sites for*, *sites for leisure*, *preserving historical sites*). It should be noted that, because these MWSs did not occur frequently in the reference corpus, they did not receive either frequency or association scores in the n-gram analysis. Our interpretation of the results is suggestive of task effects complicating the relationship between phraseological competence and general L2 proficiency. Our analysis and interpretation are descriptive and speculative, rather than conclusive, and future research could focus on this relationship to see if these results are replicated with different tasks (e.g., independent vs. integrated speaking tasks) and under different test conditions.

In answer to the second research question, regarding the relationship between use of MWSs and oral fluency, moderate correlations were observed between several n-gram factors (High-Frequency Trigram, Association Measure of MI, and Proportion, as extracted from fine-grained n-gram measures) and fluency measures (articulation rate, frequency of mid-clause and end-clause pauses, and frequency of total repair). First, the moderate positive correlation between speed fluency and high-frequency n-grams suggested that speakers with a high rate of speech produce more high-frequency combinations of words (particularly more trigrams). This finding is in line with the psycholinguistic research evidence suggesting that MWSs are stored and retrieved as individual units, allowing for information to be processed more quickly and conveniently (Ellis, 2012; Siyanova-Chanturia & Van Lancker Sidtis, 2018). We believe these findings shed further light on language processing in two ways: one at the formulation stage and the other at the articulation stage of speech production.

First, at the formulation stage, learners with a large repertoire of MWSs can retrieve longer chunks of language holistically, possibly with similar processing load to that required for retrieval of single-word items. With such retrieval efficiency, speakers might have sufficient remaining attentional resources to use for other aspects of language processing, such as syntactic and phonological encoding (Kormos, 2006). This processing efficiency might enhance speed fluency.

Second, phraseological knowledge might also influence the later stage of speech production, that is, execution of articulation (Kormos, 2006) or, more specifically, articulation of individual words (Suzuki & Kormos, 2019). After phonemic segments or lexemes are mapped onto the selected lexical items (phonological encoding), speakers execute overt production, drawing on such phonologically encoded information (Kormos, 2006; Levelt, 1989). This phonological information is hypothesized to serve as "addresses for stored phonetic syllable templates" or "motor instructions" for translating the phonologically filled frames into phonetic or articulatory programs (Levelt, 1992, p. 16). Given that MWSs, particularly the high-frequency ones, are subject to reduction of phonetic durations (e.g., deletion of [t] in *I don't know*; Bybee & Scheibman, 1999), phraseologically proficient speakers could draw on articulatory instructions stored in their lexicon and produce utterances at a faster rate in general. Conversely, speakers lacking productive knowledge of MWSs may not have access to such information about phonetic reduction, and instead they may articulate every constituent word in full form (e.g., pronouncing the sound of [t] in *I don't know*), slowing down their overall articulation rate.

Our results also suggested that use of n-grams was related to breakdown fluency in terms of mid- and end-clause pausing. For end-clause pausing, we observed a positive correlation with the MI measure, suggesting that when speakers used more strongly associated n-grams, they paused more frequently in end-clause positions. This is in line with previous research suggesting that both L1 and L2 speakers are likely to pause before producing more difficult or sophisticated single-word items indexed by frequency (de Jong, 2016). The results also suggested that the speakers who used a larger proportion of MWSs paused less frequently at mid-clause positions. This is an important finding because mid-clause pausing is believed to be linked to the formulation stage of the speech production process (Skehan, 2014). SLA researchers (e.g., Felker, Klockmann, & de Jong, 2019; Kahng, 2014) have long speculated that mid-clause pausing during L2 speech highlights the speakers' need to deal with the lexical and morphosyntactic demands of speech processing. Our findings support this view because they indicated that lack of phraseological knowledge increases the likelihood that learners will pause in an unpredictable manner. Specifically, for less proficient learners there is a risk of retrieval failure in their attempt to access and select individual words, whereas more phraseologically proficient learners are able to retrieve prefabricated chunks as a whole, providing them with "zones of safety" (Boers et al., 2006, p. 247), so that for them the risk of pausing would be more likely to be confined to the spaces in between the MWSs in their utterances (e.g., syntactic or semantic boundaries).

The results indicate that repair fluency is negatively linked with the Association Measure of MI, indicating that speakers who demonstrated more strongly associated n-grams repaired their speech less frequently. This finding implies that speakers producing MWSs with higher MI scores make fewer repetitions, reformulations, or self-corrections. It is also possible to argue that L2 learners who make more repairs draw less on formulaic language and target like MWSs. It is essential to note that the existing literature has so far documented mixed findings concerning the role of repair phenomena in L2 proficiency and oral development (Saito, Ilkan, Magne, Tran, & Suzuki, 2018; Tavakoli et al., 2020). Some researchers (e.g., Gilabert, 2007) have argued that repair behavior is more closely linked with the accuracy rather than the fluency aspect of performance, due to L2 learners' attention to well-formedness.

The qualitative analysis of the use of n-grams across proficiency levels highlighted several potentially important patterns related to the use of n-grams. First, the results underlined the important influence of task instructions on use of task-specific MWSs, supporting the well-documented learner behavior of text mining (i.e., transferring MWSs from input texts to L2 output; Boers et al., 2006; Hoang & Boers, 2016). The observed tendency of advanced speakers to use a great number of task-specific expressions accords with previous research exploring cohesion (or keyword and key-phrase overlap) between the task prompt and the speaker response and its impact on expert ratings of oral proficiency in the TOEFL-iBT integrated tasks (Crossley, Kyle, & Dascalu, 2019). In similar findings to those of the current study, Crossley et al. (2019) reported that advanced speakers used a number of key-phrases (i.e., four-word sequences from the source text) in their speech, and that expert raters judged such speech samples to be proficient.

Second, our analysis revealed that more advanced L2 speakers were more likely to avoid redundancy and repetition in their performance. Avoidance of redundancy in language use have been widely documented in previous corpus-based research (see Granger, 2018, for a review). Our results imply that avoidance of repetition and the ability to manipulate MWSs could be considered a characteristic of advanced-level L2 speakers.

Third, the qualitative analysis highlighted an interesting pattern of use of discourse markers across proficiency level: The more proficient speakers used discourse markers more frequently. This finding supports earlier research documenting an increase in the use of discourse markers in oral production when proficiency develops (Tavakoli, 2018), and highlights the significant role of discourse markers as a useful communicative device for compensating for resource deficits in L2 processing and giving a positive impression of speakers (Préfontaine & Kormos, 2016).

Limitations and Future Directions

There are several limitations of the current study worth noting for future investigation of the relationship between spoken MWSs, oral proficiency, and oral fluency. First, the study used a relatively small sample size of 56 participants' data at only four levels of oral proficiency. Replicating these findings with a larger sample size and a wider spectrum of proficiency levels would provide a more in-depth insight into the relationships between MWSs, oral proficiency, and oral fluency.

Second, the measurement of MWSs in this study was restricted to corpus frequency-based measures, disregarding other linguistic features characterizing formulaicity, such as syntactic relations (e.g., adjective + noun, adverb + verb), semantic transparency (e.g., *kick a goal* vs. *kick the bucket*), and phonological features (e.g., intonation unit).

Third, this study analyzed only two-word and three-word sequences in measuring MWSs, and therefore the result may not be generalizable to longer sequences (e.g., four- and five-word sequences). Future studies are needed to investigate longer MWSs and their relationship to oral proficiency and to oral fluency in particular.

Fourth, we based our MWS measures on n-gram tokens rather than types. Future research should investigate the differences between analyses based on n-gram tokens versus n-gram types.

Finally, like earlier lexical bundle studies (e.g., Huang, 2015), this study did not distinguish semantically incomplete n-grams (e.g., *the fact that*) from semantically complete n-grams (e.g., *in other words*). A recent psycholinguistic study conducted by Jeong and Jiang (2019) found that semantically complete units were processed differently from semantically incomplete units, suggesting that the former are more psychologically valid than the latter as holistically represented MWSs in the mental lexicon. Given this finding, future studies should consider semantic and syntactic completeness in exploring the extent to which knowledge of MWSs predicts oral proficiency.

Conclusion

In the current study we aimed to provide a fine-grained picture of the role of MWSs in oral fluency across four levels of proficiency assessed by a validated test of speaking. To the best of our knowledge, this is the first study to have investigated this relationship in a systematic manner, both quantitatively and qualitatively, across four levels of oral proficiency. The findings confirmed that different aspects of oral fluency were associated with the proportion, frequency, and association strength of the MWSs produced by the speakers. More specifically, our results indicated that (a) greater use of high-frequency MWSs in speech is linked to a faster articulation rate; (b) greater proportion of MWSs negatively correlates with the frequency of mid-clause pauses; (c) greater use of strongly associated MWSs negatively correlates with repair phenomena; and (d) greater use of strongly associated MWSs positively correlates with the frequency of end-clause pauses.

Our qualitative analysis suggested that some of the MWSs were borrowed from task instructions and were used frequently, perhaps because they were central to task completion. The qualitative findings also suggested that low oral proficiency speakers relied more on verbatim repetition of the borrowed MWSs, whereas high oral proficiency speakers were more likely to avoid repetition and to use these MWSs more creatively.

In their totality, these findings are important because they show how the lexical demands of producing L2 speech relate to oral fluency, and they suggest that this relationship may vary at different levels of oral proficiency and may be affected by task rubrics. The results are also important for language testing organizations in informing the development of rating descriptors and rating scales to evaluate use of spoken formulaic language.

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Notes

1 A third rater was invited to rate the samples independently if the first and second cannot agree.

2 We expected to observe strong and comparable loadings of bigram and trigram frequency indices on a Frequency factor. However, because bigram and trigram frequency did not load onto the same factor, we decided to label Factor 1 in terms of trigram frequency rather than bigram and trigram frequency.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1. Table 8. Results of Comparison of N-gram Frequency, Proportion, and Association Scores across Four Proficiency Levels
Appendix S2. Exact *p*-Values for Each Comparison of N-Gram Measures Across Four Proficiency Levels
Appendix S3. Table 9. The Most Frequent 20 Bigrams and Trigrams Used by Low B1 Speakers
Appendix S4. Table 10. The Most Frequent 20 Bigrams and Trigrams Used by High B1 Speakers
Appendix S5. Table 11. The Most Frequent 20 Bigrams and Trigrams Used by B2 Speakers
Appendix S6. Table 12. The Most Frequent 20 Bigrams and Trigrams Used by C1 Speakers

Appendix: Accessible Summary (also publicly available at https://oasis-database.org)

Fluency and Multiword Sequences Across Different Levels of Oral Proficiency

What This Research Was About and Why It Is Important

Multiword sequences (MWSs) are combinations of words that appear together very frequently in a language. They include various forms of lexical strings, such as idioms (e.g., *kick the bucket, spill the beans*), restricted collocations (e.g., *heavy traffic, blow a fuse*), phrasal verbs (e.g., *hung up, call off*), binomials (e.g., *bride and groom*, *ladies and gentlemen*), proverbs (e.g., *birds of a feather* *flock together*), and lexical bundles or n-grams (e.g., *and so on*, *one of the, I don't know*).

It is important to understand how such MSWs are used as second language (L2) proficiency improves, and to what extent these sequences contribute to oral fluency (that is the flow, smoothness and continuity of speech). For example, how MWSs relate to oral fluency and proficiency is important for assessing language proficiency. This study found that a greater use of MWSs was associated with higher oral proficiency scores and also with some characteristics of fluency (speech rate and fewer pauses mid-clause) but not others (pauses in between clauses).

What the Researchers Did

- The researchers recorded and transcribed the speech of 56 learners doing the oral tasks in the TEEP across four levels of proficiency (Low B1 to C1 of the Common European Framework of Reference).
- MWSs were identified, by ensuring that all MSWs were among 30,000 most frequent MWSs in a native speaker corpus.
- Very precise measures for oral fluency (speed, breakdown, and repair) were calculated, using digital software (PRAAT).
- The MWSs were scored for frequency, proportion and association strength (how likely the words are to occur together), using digital software (TAALES).
- The researchers examined (a) the amount and type of use of MWSs at the four different proficiency levels and (b) the strength of the associations (correlations) between MWSs and oral fluency

What the Researchers Found

The researchers found that:

- more frequent use of these two- and three-word MWSs was associated with higher proficiency levels for some of the MWSs measures;
- speakers who used more high-frequency MWSs were faster at talking (in their articulation rate);
- speakers who used more MWSs paused less frequently in mid-clause positions;
- using strongly associated MWSs (words that are highly likely to occur together) was linked to having fewer repairs (self-corrections) and more pauses at the end of clauses;

An exploratory descriptive analysis suggested that:

- some MWSs were borrowed from the task instructions and were used frequently perhaps because they were central to task completion
- while lower proficiency speakers relied more on verbatim repetition of the MWSs borrowed from the task, high-proficiency speakers seemed more likely to avoid exact repetition form the instruction and use MWSs more creatively.

Things to Consider

- Future research will need to look at a wider range of proficiency levels.
- A larger sample size will also help ensure generalizability of the results in future studies.
- This study explored only two- and three-word sequences (bigrams and trigrams); more research is needed to explore the relationship between longer sequences and proficiency.

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